

EDWARD S. RICEMEYER

(né Rice)
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EDUCATION

- 2025- Habilitand in Genomics
 Faculty of Veterinary Medicine, Ludwig-Maximilians-Universität
- 2018 Ph.D. in Bioinformatics
 Department of Biomolecular Engineering, University of California, Santa Cruz
- 2012 A.B. in Computer Science
 Department of Computer Science, Brown University

PROFESSIONAL APPOINTMENTS

- 2026- Senior Research Scientist
 Faculty of Veterinary Medicine, Ludwig-Maximilians-Universität
- 2019-26 Senior Research Scientist
 Bond Life Sciences Center, University of Missouri
- 2018-19 Postdoctoral Research Associate
 University of Nebraska - Lincoln

PUBLICATIONS

(note birth name Rice)
* co-first authorship

Preprints and submitted manuscripts

Edward S. Ricemeyer*, Kathryn Gallman*, Maggs X*, Yulia Nussbaum, Rachel A. Carroll, Robert Peuß, Nicolas Rohner, Alex C. Keene & Wesley C. Warren (in review, *Molecular Biology and Evolution*). “Cave adaptation drives coordinated transcriptional remodeling across diverse cell types in the brain of a teleost fish.” Preprint: *bioRxiv* 2026.06.19.733352.

Kathryn Gallman, **Edward S. Ricemeyer**, Aakriti Rastogi, Maggs X, Emilio Méndez Scolari, Erik R. Duboue, Nicolas Rohner, Harini Iyer, Wesley C. Warren, & Alex C. Keene (in review, *Communications Biology*). “A comparative brain atlas in the Mexican cavefish identifies widespread changes in cellular composition and gene expression.” Preprint: *bioRxiv* 2026.05.12.723532.

Natalie Choi, **Edward S. Ricemeyer**, Maggs X, Zicong Zhang, Masato Nikaido, Wesley C. Warren, & Masato Yoshizawa (in revision). “Novel mechanisms of chemosensory adaptation to the cave environment.” Preprint: *bioRxiv* 2026.04.03.716434.

Lauren M. Hennelly, **Edward S. Ricemeyer**, Katia Bougiouri, Lachie Scarsbrook, Melanie Fillios, Kylie M. Cairns, Nuno F. G. Martins, Renzo Balboa, Cindy G Satander, Philippe Gaubert, Eli K. Rueness, Jazlyn A. Mooney, Ellie E. Armstrong, Benjamin N. Sacks, Niamh Quinn, Sarah B. Carey, Alex Harkess, Camilla Scharff-Olsen, Liam T. Lanigan, M. Thomas P. Gilbert, Kathryn Lord, Greger Larson, Nucharin Songsasen, Klaus-Peter Koepfli, Shyam Gopalakrishnan, Mikkel-Holger S. Sinding, Laurent A. F. Frantz (in revision). “Tracking the evolutionary history of aseasonal reproduction in dogs.”

Published and in press

Jian-Hai Chen, Xiaoyong Du, Zhuqing Zheng, Neena Amatya Gorkhali, Changyi Xiao, Shilin Zhu, Jie Zhong, Liangliang Fu, Jingjin Li, Zhuofan Yan, **Edward S. Ricemeyer**, Fan Jiang, Thuy Nhien Thi Tran, Pan Ni, Nicholas VanKuren, Guanglin He, Yalan Yang, Supamit Mekchay, San San Hmwe, Mikhail Sablin, Nikica Šprem, Le Thi Thuy, Abdullah Omar, M. O. Faruque, Ali Akbar Bhuiyan, G. L. L. Pradeepa Silva, Amali Malshani Samaraweera, Debabrata Phukon, Mukesh Thakur, Saroj Sapkota, S. M. Farhad Vahidi, Faiz-ul Hussan, Hidayat Ashari, Bayu Dewantoro Putro Soewandi, Ivan Jakovlić, Kevin G. Daly, Ming-Shan Wang, Xiaolei Liu, Malavanh Chittavong, Daniel Falush, Jeffrey Spence, Richard Bishop, David W. G. Stanton, Xuewen Xu, Xinyun Li, Matthew W. Hahn, Lingzhao Fang, Laurent A. F. Frantz, Greger Larson, Jianlin Han, Shuhong Zhao (2026). “Ancient introgression drives wild boar expansion and phenotypic diversification of domestic pigs.” *Science*, 393(6818), 1335-1341.

Emma Y. Roback*, Maggs X*, **Edward S. Ricemeyer**, Adam Warlen, Rachel A. Carroll, Christine G. Elsik, Alex C. Keene, Nicolas Rohner, Suzanne E. McGaugh & Wesley C. Warren (2026). “A pangenome framework uncovers the role of deletions in repeated evolution of cave-derived traits.” *Genome Research*, 36(9), 1921-1939.

Edward S. Ricemeyer*, Nathan K. Schaefer*, Kang Du, Irene da Cruz, Susanne Kneitz, Rafael D. Acemel, Dario G. Lupiáñez, Rachel A. Carroll, Rosie Drinkwater, Manfred Scharl & Wesley C. Warren (2026). “Gene conversion empowers natural selection in a clonal fish species.” *Nature*, 652(8109), 398-404.

Isabell Naumann, **Edward S. Ricemeyer**, Daniel Elleder, Jiri Plachy, Dominik von La Roche, Kim Vučinić, Thomas W. Göbel, Bernd Kaspers, Simon P. Früh & Sonja Härtle (2025). “Beyond FoxP3 – Identification of a chicken regulatory T cell signature.” *European Journal of Immunology*, 55(12), e70106.

Rachel A. Carroll, **Edward S. Ricemeyer**, LaDeana W. Hillier, Jeffrey M. DaCosta, Ekaterina Osipova, Sara Smith, Gabriel A. Jamie, Juan G. Martinez, Mercedes Molina-Morales, Tomas Marques-Bonet, Joseph D. Manthey, Diana Haddad, Matthew J. Fuxjager, Kathleen S. Lynch, Jonathan M. D. Wood, Erich Jarvis, Patrick Masterson, Francoise Thibaud-Nissen, Mark E.

- Hauber, Claire N. Spottiswoode, Timothy B. Sackton, Christopher N. Balakrishnan, Michael D. Sorenson & Wesley C. Warren (2025). “Genomic resources for comparative analyses of obligate avian brood parasitism.” *Journal of Heredity*, esaf087.
- Yulia I. Nussbaum, Jeffrey N. Bryan, Guangfu Li, Alana Rodney, William T. N. Culp, Peter J. Dickinson, Anna Koutoulas, Kate Megquier, Hong An, Skyler T. Kramer, **Edward S. Ricemeyer**, Jonathan B. Mitchem & Wesley C. Warren (2025). “Comparative multi-omics evaluation of the tumor microenvironment in canine and human hepatocellular carcinoma.” *Veterinary Oncology*, 2, 25.
- Leah K. Treffer, Renae L. Schroeder, **Edward S. Ricemeyer**, Ted Kalbfleisch, Anna M. Fuller, Jessica L. Petersen (2025). “Mitochondrial introgression with potential functional effects in North American yak.” *Ecology and Evolution*, 15(10), e72362.
- Edward S. Ricemeyer**, Rachel A. Carroll & Wesley C. Warren (2025). “Agptools: a utility suite for editing genome assemblies.” *Bioinformatics*, 41(7), btaf388.
- Wesley C. Warren, **Edward S. Rice**, Maggs X, Emma Roback, Alex Keene, Fergal Martin, Denye Ogeh, Leanne Haggerty, Rachel A Carroll, Suzanne McGaugh & Nicolas Rohner (2024). “*Astyanax mexicanus* surface and cavefish chromosome-scale assemblies for trait variation discovery.” *G3: Genes, Genomes, Genetics*, 14(8).
- Rachel A. Carroll, **Edward S. Rice**, William J. Murphy, Leslie A. Lyons, Francoise Thibaud-Nissen, Lyndon M. Coghill, William F. Swanson, Karen A. Terio, Tyler Boyd & Wesley C. Warren (2024). “A chromosome-scale fishing cat reference genome for the evaluation of potential germline risk variants.” *Scientific Reports*, 14, 8073.
- Edward S. Rice**, Anton Alberdi, James Alfieri, Giridhar Athrey, Jennifer R. Balacco, Philippe Bardou, Heath Blackmon, Mathieu Charles, Hans H. Cheng, Olivier Fedrigo, Steven R. Fiddaman, Giulio Formenti, Laurent A. F. Frantz, M. Thomas P. Gilbert, Cari J. Hearn, Erich D. Jarvis, Christophe Klopp, Sofia Marcos, Andrew S. Mason, Deborah Velez-Irizarry, Luohao Xu & Wesley C. Warren (2023). “A pangenome graph reference of 30 chicken genomes allows genotyping of large and complex structural variants.” *BMC Biology*, 21, 267.
- Kevin R. Bredemeyer, LaDeana Hillier, Andrew J. Harris, Graham M. Hughes, Nicole M. Foley, Colleen Lawless, Rachel A. Carroll, Jessica M. Storer, Mark A. Batzer, **Edward S. Rice**, Brian W. Davis, Terje Raudsepp, Stephen J. O’Brien, Leslie A. Lyons, Wesley C. Warren & William J. Murphy (2023) “Single-haplotype comparative genomics reveals the role of structural innovations during cat evolution.” *Nature Genetics*, 55, 1953-1963.
- Yuan Lu*, **Edward Rice***, Kang Du, Susanne Kneitz, Magali Naville, Corentin Dechaud, Jean-Nicolas Volff, Mikki Boswell, William Boswell, LaDeana Hillier, Chad Tomlinson, Kremitzki Milin, Ronald B. Walter, Manfred Schartl & Wesley C. Warren (2023). “High resolution genomes of multiple *Xiphophorus* species provide new insight into microevolution, hybrid incompatibility, epistasis and disease.” *Genome Research*, 33, 557-571.

Jonathan B. Puritz, Ximing Guo, Matthew Hare, Yan He, LaDeana W. Hillier, Shubo Jin, Ming Liu, Katie E. Lotterhos, Pat Minx, Tejashree Modak, Dina Proestou, **Edward S. Rice**, Chad Tomlinson, Wesley C. Warren, Erin Witkop, Honggang Zhao & Marta Gomez-Chiarri (2023). “A second unveiling: haplotig masking of the eastern oyster genome improves population-level inference.” *Molecular Ecology Resources*, 2023(00), 1-18.

Wesley C. Warren, **Edward S. Rice**, Ashley Meyer, Cari J. Hearn, Alec Steep, Henry D. Hunt, Melissa S. Monson, Susan J. Lamont & Hans H. Cheng (2023). “The immune cell landscape and response of Marek’s disease resistant and susceptible chickens infected with Marek’s disease virus.” *Scientific Reports*, 13:5355.

Wesley Warren, Olivier Fedrigo, Alan Tracey, Andrew S. Mason, Giulio Formenti, Perini Francesco, Zhou Wu, Terence Murphy, Valerie Schneider, Kyle Stiers, **Edward S. Rice**, Lyndon M. Coghill, Nick Anthony, Ron Okimoto, Rachel Carroll, Jacquelyn Mountcastle, Jennifer Balacco, Bettina Haase, Chentao Yang, Guojie Zhang, Jacqueline Smith, Yvonne Drechsler, Hans H. Cheng, Kerstin Howe & Erich D Jarvis (2023). “Multiple chicken (*Gallus gallus*) genome references to advance genetic variation studies.” *Cytogenetic and Genome Research*, 162, 8-9.

Helen G. Potts, Madeleine E. Lemieux, **Edward S. Rice**, Wesley Warren, Robin P. Choudhury & Mathilda T. M. Mommersteeg (2022). “Discordant genome assemblies drastically alter the interpretation of single cell RNA sequencing data which can be mitigated by a novel integration method.” *Cells*, 11, 608.

Richard J. Edwards, Matt A. Field, James M. Ferguson, Olga Dudchenko, Jens Keilwagen, Benjamin D. Rosen, Gary S. Johnson, **Edward S. Rice**, La Deanna Hillier, Jillian M. Hammond, Samuel G. Towarnicki, Arina Omer, Ruqayya Khan, Ksenia Skvortsova, Ozren Bogdanovic, Robert A. Zammit, Erez Lieberman Aiden, Wesley C. Warren & J. William O. Ballard (2021). “Chromosome-length genome assembly and structural variations of the primal Basenji dog (*Canis lupus familiaris*) genome.” *BMC Genomics*, 22, 188.

Wesley C. Warren, Tyler E. Boggs, Richard Borowsky, Brian M. Carlson, Estephany Ferrufino, Joshua B. Gross, LaDeana Hillier, Zhilian Hu, Alex C. Keene, Alexander Kenzior, Johanna E. Kowalko, Chad Tomlinson, Milinn Kremitzki, Madeleine E. Lemieux, Tina Graves-Lindsay, Suzanne E. McGaugh, Jeffrey T. Miller, Mathilda T. M. Mommersteeg, Rachel L. Moran, Robert Peuß, **Edward S. Rice**, Misty R. Riddle, Itzel Sifuentes-Romero, Bethany A. Stanhope, Clifford J. Tabin, Sunishka Thakur, Yoshiyuki Yamamoto & Nicolas Rohner (2021). “A chromosome-level genome of *Astyanax mexicanus* surface fish for comparing population-specific genetic differences contributing to trait evolution.” *Nature Communications*, 12, 1447.

Cleber A. Trujillo, **Edward S. Rice**, Nathan K. Schaefer, Isaac A. Chaim, Emily C. Wheeler, Assael A. Madrigal, Justin Buchanan, Sebastian Preissl, Allen Wang, Priscilla D. Negraes, Ryan A. Szeto, Roberto H. Herai, Alik Huseynov, Mariana S. A. Ferraz, Fernando S. Borges,

Alexandre H. Kihara, Ashley Byrne, Maximillian Marin, Christopher Vollmers, Angela N. Brooks, Jonathan D. Lautz, Katerina Semendeferi, Beth Shapiro, Gene W. Yeo, Stephen E. P. Smith, Richard E. Green & Alysson R. Muotri (2021). “Reintroduction of the archaic variant of NOVA1 in cortical organoids alters neurodevelopment.” *Science*, 371(6530), eaax2537.

Abdelnaby Khalyfa, Wesley Warren, Jorge Andrade, Christopher A. Bottoms, **Edward S. Rice**, Rene Cortese, Leila Kheirandish-Gozal & David Gozal (2021). “Transcriptomic Changes of Murine Visceral Fat Exposed to Intermittent Hypoxia at Single Cell Resolution.” *Int. J. Mol. Sci*, 22, 261.

Edward S. Rice, Sergey Koren, Arang Rhie, Michael P. Heaton, Theodore S. Kalbfleisch, Timothy Hardy, Peter H. Hackett, Derek M. Bickhart, Benjamin D. Rosen, Brian Vander Ley, Nicholas W. Maurer, Richard E. Green, Adam M. Phillippy, Jessica L. Petersen & Timothy P. L. Smith (2020). “Continuous chromosome-scale haplotypes assembled from a single interspecies F1 hybrid of yak and cattle.” *GigaScience*, 9(4), giaa029.

Edward S. Rice & Richard E. Green (2019). “New approaches for genome assembly and scaffolding.” *Annual Review of Animal Biosciences*, 7, 17-40.

Theodore S. Kalbfleisch*, **Edward S. Rice***, Michael S. DePriest Jr., Brian P. Walenz, Matthew S. Hestand, Joris R. Vermeesch, Brendan L. O'Connell, Ian T. Fiddes, Alisa O. Vershinina, Nedda F. Saremi, Jessica L. Petersen, Carrie J. Finno, Rebecca R. Bellone, Molly E. McCue, Samantha A. Brooks, Ernest Bailey, Ludovic Orlando, Richard E. Green, Donald C. Miller, Douglas F. Antczak & James N. MacLeod (2018). “Improved reference genome for the domestic horse increases assembly contiguity and composition.” *Communications Biology*, 1(1), 197.

Gemma G. R. Murray, André E. R. Soares, Ben J. Novak, Nathan K. Schaefer, James A. Cahill, Allan J. Baker, John R. Demboski, Andrew Doll, Rute R. Da Fonseca, Tara L. Fulton, M. Thomas P. Gilbert, Peter D. Heintzman, Brandon Letts, George McIntosh, Brendan L. O'Connell, Mark Peck, Marie-Lorraine Pipes, **Edward S. Rice**, Kathryn M. Santos, A. Gregory Sohrweide, Samuel H. Vohr, Russell B. Corbett-Detig, Richard E. Green & Beth Shapiro (2017). “Natural selection shaped the rise and fall of passenger pigeon genomic diversity.” *Science*, 358(6365), 951-954.

Edward S. Rice, Satomi Kohno, John St. John, Son Pham, Jonathan Howard, Liana F. Lareau, Brendan L. O'Connell, Glenn Hickey, Joel Armstrong, Alden Deran, Ian Fiddes, Roy N. Platt II, Cathy Gresham, Fiona McCarthy, Colin Kern, David Haan, Tan Phan, Carl Schmidt, Jeremy R. Sanford, David A. Ray, Benedict Paten, Louis J. Guillette Jr. & Richard E. Green (2017). “Improved genome assembly of American alligator genome reveals conserved architecture of estrogen signaling.” *Genome Research*, 27(5), 686-696.

Raudsepp, T., Gustafson-Seabury, A., Durkin, K., Wagner, M. L., Goh, G., Seabury, C. M., Brinkmeyer-Langford, C., Lee, E. J., Agarwala, R., **Rice, E. S.**, Schäffer, A. A., Skow, L. C., Tozaki, T., Yasue, H., Penedo, M. C., Lyons, L. A., Khazanehdari, K. A., Binns, M. M.,

MacLeod, J. N., Distl, O., Guérin, G., Leeb, T., Mickelson, J. R., & Chowdhary, B. P. (2008). "A 4,103 marker integrated physical and comparative map of the horse genome." *Cytogenetic and Genome Research*, 122(1), 28-36.

Alejandro A. Schäffer, **Edward S. Rice**, William Cook & Richa Agarwala (2007). "rh_tsp_map 3.0: end-to-end radiation hybrid mapping with improved speed and quality control." *Bioinformatics*, 23(9), 1156-1158.

AWARDS

- 2026 Deutsche Forschungsgemeinschaft (DFG) Individual Research Grant
"A pangenome of worldwide chicken diversity to discover structural variants impacting health, sustainability, breeding, and domestication." € 495,119
- 2024 SuperMUC-NG Regular Project Grant, Leibniz Supercomputing Center
"Leveraging genome assemblies to establish the impact of structural variation on domestic animals," 6M CPU-hours (ca. € 375,000 value)
- 2018 Horse Genome Coordinators' Travel Award, \$1,000
- 2017 ARCS Foundation Scholarship, \$11,000
- 2017 Baskin Dissertation Year Fellowship, \$39,267
- 2015 Genome 10K Conference Presenter's Fellowship
- 2013 UC Santa Cruz Chancellor's Fellowship, \$55,278
- 2012 National Institutes of Health Intramural Research Training Award, \$30,000
- 2009 Karen T. Romer Undergraduate Teaching and Research Award, \$3,000

INVITED SEMINARS

UC Davis; University of Münster

CONFERENCE TALKS

(selection)

- 2026 Astyanax International Meeting, Mexico City, Mexico
Conference of Poeciliid and Goodeid Biologists, Padua, Italy
German Zoological Society, Münster, Germany
- 2025 Plant and Animal Genomes, San Diego, CA, USA
European Cavefish Meeting, Wettringen, Germany
- 2024 Astyanax International Meeting, Fort Lauderdale, FL, USA
AnimalFarm, Naples, Italy
- 2023 Plant and Animal Genomes, San Diego, CA, USA
EvoGenMunich, Munich, Germany
- 2022 Astyanax International Meeting, San Antonio, TX, USA
Plant and Animal Genomes, online
- 2019 Advances in Genome Biology and Technology, Marco Island, FL, USA

2018 Plant and Animal Genomes, San Diego, CA, USA

TEACHING

As instructor

Introduction to Programming, Ludwig-Maximilians-Universität (1 term, 2025)

As guest lecturer

Introduction to Evolutionary Biology, Ludwig-Maximilians-Universität (4 terms, 2023-2026)

As teaching assistant

The Human Genome, UC Santa Cruz (4 quarters, Fall 2015 - Spring 2017)

Technical Writing and Communication in Computer Science, UC Santa Cruz (Winter 2017)

Bioinformatics Models and Algorithms, UC Santa Cruz (Fall 2016)

ACADEMIC SERVICE

Journal Reviews

Agriculture Communications; Animal Genetics; BMC Genomics; British Poultry Science; Genetics Selection Evolution; GigaScience; Heredity; Integrative & Comparative Biology; Journal of Experimental Zoology Part B; Journal of Heredity; Molecular Biology and Evolution; Molecular Ecology Resources; Nature; Nature Communications; Scientific Data

SKILLS

Technical

15+ years programming experience in Python, R, C/C++ (github.com/esrice)

Proficient with Unix environments and standard bioinformatics databases, file formats, and software

Languages

English (native), German (basic proficiency)